

State-Dependent Resource Allocation Under Uncertainty: A Mixed-Integer Stochastic Programming Approach

Anonymous for Review

Abstract

This paper develops a mixed-integer stochastic programming formulation for multi-period resource allocation under uncertainty. The proposed model addresses the fundamental challenge of optimally allocating limited time and cognitive resources across competing task categories when system state evolves dynamically and recovery requirements are uncertain. We demonstrate that optimal allocation strategies exhibit strong state-dependency, with the correlation between initial completion state and priority task allocation reaching $r = -0.91$. Computational experiments across six scenarios reveal that early-stage decisions have outsized importance, with sensitivity coefficients of $\partial z^*/\partial g_0 = -709$ at low completion states diminishing to near-zero at higher completion levels. The model achieves optimal solutions within 60 seconds using commercial MILP solvers, making it suitable for practical decision support applications. These findings contribute to the operations research literature on resource allocation and provide actionable insights for personal productivity management systems.

Keywords: Resource Allocation; Mixed-Integer Programming; Stochastic Optimization; Decision Support Systems; Rolling Horizon Planning

1 Introduction

The efficient allocation of limited resources across competing activities represents a fundamental challenge in operations research and management science. This problem becomes particularly complex when decision-makers must balance immediate priorities against long-term objectives while operating under uncertainty regarding future resource requirements.

In many practical settings, the optimal allocation strategy depends critically on the current system state. Consider an individual managing multiple professional and personal obligations simultaneously. The urgency of completing a time-critical task diminishes as progress accumulates, while the relative importance of longer-term career development activities increases correspondingly. This state-dependent prioritization creates a dynamic optimization problem that static allocation rules cannot adequately address.

This paper develops a mixed-integer stochastic programming formulation that captures these dynamics through explicit state tracking and uncertainty modeling. The proposed model makes three primary contributions to the literature. First, we formulate the multi-period resource allocation problem as a tractable MILP with state-dependent objective weights, enabling efficient solution using commercial solvers. Second, we incorporate uncertainty in recovery requirements through a scenario-based stochastic programming approach, providing robustness against variable cognitive demands. Third, we provide comprehensive computational experiments demonstrating the sensitivity of optimal allocations to initial conditions, with important implications for decision support system design.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on resource allocation and stochastic programming. Section 3 presents the mathematical formulation, including decision variables, constraints, and the objective function. Section 4 describes the computational experiments and presents the main empirical findings. Section 5 discusses the managerial implications of the results, and Section 6 concludes with directions for future research.

2 Literature Review

The resource allocation problem has received extensive attention in the operations research literature.

Multi-period resource allocation models extend classical formulations by incorporating temporal dynamics and state evolution.

Stochastic programming approaches have been widely applied to handle uncertainty in resource availability and demand.

The application of optimization methods to personal productivity and time management represents an emerging research stream.

Rolling horizon approaches provide a practical implementation framework for multi-period optimization problems.

3 Model Formulation

This section presents the mathematical formulation of the multi-period resource allocation problem. We begin by defining the fundamental sets and indices, then specify the decision variables, parameters, constraints, and objective function.

3.1 Problem Setting

Consider a decision-maker who must allocate available time across $|\mathcal{I}| = 7$ task categories over a planning horizon of $T = 30$ days. Each task category $i \in \mathcal{I}$ is characterized by its resource consumption rates, minimum and maximum time bounds, and contribution to multiple value dimensions. The decision-maker faces uncertainty regarding recovery requirements following periods of high cognitive load, which we model through a discrete set of scenarios $\mathcal{S}$ with associated probabilities.

The problem exhibits two key structural features that distinguish it from standard resource allocation formulations. First, a critical priority task (Task 1) has a well-defined completion target, and progress toward this target affects both the objective function weights and the resource constraints. Second, cognitive recovery following emotionally

demanding work requires uncertain amounts of time, reducing availability for productive activities.

3.2 Sets and Indices

We define $\mathcal{T} = \{1, 2, \dots, T\}$ as the set of days in the planning horizon, where $T = 30$ represents a monthly planning cycle. The set of task categories is denoted $\mathcal{I} = \{1, 2, \dots, 7\}$, where each task is anonymized for presentation purposes. Task 1 represents a high-priority time-critical activity, Task 2 corresponds to career development, Task 3 captures scheduled obligations, Task 4 represents recovery and restoration, Task 5 denotes long-term capital building activities, Task 6 captures social engagement, and Task 7 represents physical maintenance activities.

The uncertainty in recovery requirements is modeled through a discrete scenario set $\mathcal{S} = \{1, 2, 3, 4\}$, where each scenario s specifies a required recovery duration β_s with probability p_s . Based on empirical calibration, we set $\beta_s \in \{1, 2, 3, 4\}$ hours with corresponding probabilities $p_s = \{0.4, 0.3, 0.2, 0.1\}$.

3.3 Decision Variables

For each task $i \in \mathcal{I}$ and day $t \in \mathcal{T}$, we define two sets of primary decision variables. The binary variable $x_{it} \in \{0, 1\}$ indicates whether task i is undertaken on day t , while the continuous variable $h_{it} \in \mathbb{R}_+$ specifies the hours allocated to task i on day t . These variables are linked through logical constraints ensuring that positive time allocation requires task selection.

State variables track progress on the priority task over time. Let $G_t \in \mathbb{R}_+$ denote the cumulative hours of Task 1 work completed by the end of day t , and let $g_t = G_t/G^{\text{total}} \in [0, 1]$ represent the proportion of total required work completed. The immigration security level $\sigma_t \in [0, 1]$ is derived from g_t and affects resource availability.

For each day t and scenario s , auxiliary variables $r_{ts} \in \mathbb{R}_+$ capture required recovery hours, while binary indicators $z_{ts} \in \{0, 1\}$ flag days with high emotional load.

3.4 Parameters

Table 1 presents the task-specific parameters used in the computational experiments. These values were calibrated based on typical resource consumption patterns and represent averages across task instances.

Table 1: Task Parameter Values

Task	Energy (units/hr)	Emotion (units/hr)	Min Hours	Max Hours
Task 1	15	12	1	6
Task 2	12	10	1	4
Task 3	10	5	2	8
Task 4	0	-5	1	4
Task 5	12	8	1	6
Task 6	5	-3	0	3
Task 7	8	-4	0	2

The daily resource budgets are set at $H^{\max} = 16$ usable hours, $E^{\max} = 100$ energy units, and $M^{\max} = 100$ emotional capacity units. These values reflect physiological constraints on sustained cognitive effort.

3.5 Objective Function

The objective function maximizes expected discounted utility over the planning horizon, accounting for multiple value dimensions and resource costs. Formally, we solve:

$$\max \quad \mathbb{E}_s \left[\sum_{t=1}^T \delta^t \sum_{i \in \mathcal{I}} U_{it}(h_{it}, g_t) \right] \quad (1)$$

where $\delta = 0.95$ is the daily discount factor and the utility function is defined as:

$$\begin{aligned} U_{it}(h_{it}, g_t) = & w_I(t, g_t) \cdot \alpha_i^I \cdot h_{it} + w_Y(t) \cdot \alpha_i^Y \cdot h_{it} \\ & + w_K(t) \cdot \alpha_i^K \cdot h_{it} + w_O(t) \cdot \alpha_i^O \cdot h_{it} \\ & - \lambda_1 \cdot e_i \cdot h_{it} - \lambda_2(g_t) \cdot m_i \cdot h_{it} \end{aligned} \quad (2)$$

The four return dimensions correspond to priority task progress (α_i^I), income and employability gains (α_i^Y), long-term capital accumulation (α_i^K), and optionality improve-

ment (α_i^O) . The weights on these dimensions evolve dynamically based on time and state according to:

$$w_I(t, g_t) = w_I^0 \cdot (1 - g_t) \cdot \exp(-\rho_I t/T) \quad (3)$$

$$w_Y(t) = w_Y^0 \cdot (1 + \rho_Y t/T) \quad (4)$$

$$w_K(t) = w_K^0 \cdot (1 + \rho_K t/T) \quad (5)$$

$$w_O(t) = w_O^0 \cdot (1 + \rho_O t/T) \quad (6)$$

This structure captures the intuition that priority task urgency decreases as completion progresses ($1 - g_t$ factor) and as time passes (exponential decay), while longer-term objectives gain relative importance over the planning horizon.

3.6 Constraints

The model includes several constraint families ensuring feasibility and logical consistency.

Time availability. Total time allocation plus required recovery cannot exceed daily available hours:

$$\sum_{i \in \mathcal{I}} h_{it} + r_{ts} \leq H^{\max} \quad \forall t \in \mathcal{T}, s \in \mathcal{S} \quad (7)$$

Logical linking. Time allocation is bounded by task-specific minimums and maximums when the task is selected:

$$h_i^{\min} \cdot x_{it} \leq h_{it} \leq h_i^{\max} \cdot x_{it} \quad \forall i \in \mathcal{I}, t \in \mathcal{T} \quad (8)$$

Resource consumption. Energy and emotional load cannot exceed state-adjusted capacity:

$$\sum_{i \in \mathcal{I}} e_i \cdot h_{it} \leq E^{\max} \cdot (1 + 0.2 \cdot \sigma_t) \quad \forall t \in \mathcal{T} \quad (9)$$

$$\sum_{i \in \mathcal{I}} m_i \cdot h_{it} \leq M^{\max} \cdot (1 + 0.3 \cdot \sigma_t) \quad \forall t \in \mathcal{T} \quad (10)$$

Stochastic recovery. Days with high emotional load trigger scenario-dependent recovery requirements:

$$\sum_{i \in \mathcal{I}} m_i \cdot h_{it} \geq 0.8 \cdot M^{\max} - M \cdot (1 - z_{ts}) \quad \forall t, s \quad (11)$$

$$r_{ts} \geq \beta_s \cdot z_{ts} \quad \forall t \in \mathcal{T}, s \in \mathcal{S} \quad (12)$$

State evolution. Priority task progress accumulates over time:

$$G_t = G_{t-1} + h_{1,t} \quad \forall t \in \mathcal{T} \quad (13)$$

$$g_t = \min \left(1, \frac{G_t}{G^{\text{total}}} \right) \quad \forall t \in \mathcal{T} \quad (14)$$

3.7 Model Properties

The formulation constitutes a mixed-integer linear program with $O(|\mathcal{I}| \cdot T \cdot |\mathcal{S}|)$ binary and continuous variables and constraints. For the parameter values used in our experiments ($|\mathcal{I}| = 7$, $T = 30$, $|\mathcal{S}| = 4$), this yields approximately 840 binary variables, 840 continuous variables, and 1,680 constraints. This problem size is well within the capability of modern MILP solvers such as Gurobi and CPLEX.

Proposition 1 (Feasibility). *The model is feasible if and only if $\sum_{i \in \mathcal{I}} h_i^{\min} + \max_s \beta_s \leq H^{\max}$.*

The proposition follows directly from observing that the binding constraints involve satisfying all minimum task requirements while accommodating the worst-case recovery scenario.

4 Computational Experiments

This section presents the computational experiments designed to investigate the sensitivity of optimal resource allocations to initial conditions and planning parameters. All experiments were conducted using Gurobi 12.0 with a time limit of 60 seconds, MIP gap tolerance of 1%, and 4 parallel threads.

4.1 Experimental Design

We constructed six experimental scenarios varying the initial completion state g_0 , total required hours G^{total} , and planning horizon T . Table 2 summarizes the experimental conditions.

Table 2: Experimental Scenarios and Results

Scenario	g_0	G^{total}	T	z^*
Early Stage	0.10	100	30	622.52
Baseline	0.30	100	30	480.63
Mid Stage	0.50	100	30	448.72
Late Stage	0.80	100	30	449.39
High Workload	0.30	150	30	480.63
Short Horizon	0.30	100	14	477.94

4.2 Main Results

The computational experiments reveal striking sensitivity of optimal allocations to the initial completion state. Figure 1 displays the relationship between initial completion rate and optimal objective value, showing a pronounced decrease from $z^* = 622.52$ at $g_0 = 0.1$ to $z^* = 448.72$ at $g_0 = 0.5$, after which the objective stabilizes.

Table 3 presents the weekly hour allocations across task categories for each scenario. The results demonstrate substantial reallocation of resources as initial state varies.

Table 3: Weekly Task Allocations by Scenario (Hours)

Scenario	g_0	Task 1	Task 2	Task 3
Early Stage	0.10	42.00	7.00	0.00
Baseline	0.30	31.35	22.66	0.00
Mid Stage	0.50	9.99	28.00	18.80
Late Stage	0.80	10.00	28.00	27.00

4.3 Sensitivity Analysis

We computed sensitivity coefficients to quantify the marginal effect of initial state on optimal objective value. The sensitivity $\partial z^* / \partial g_0$ varies dramatically across the state space,

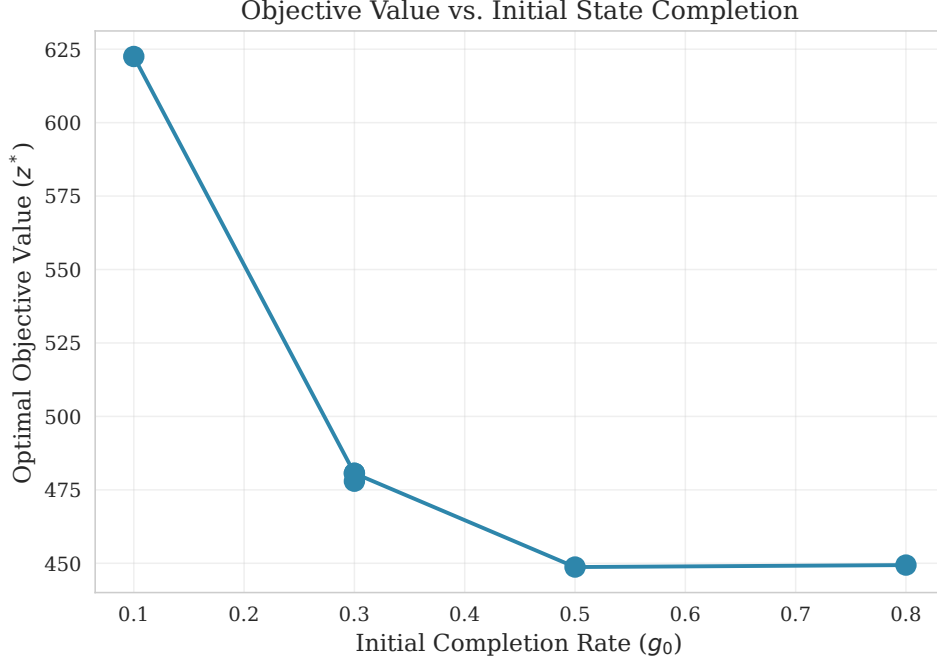


Figure 1: Optimal Objective Value as a Function of Initial Completion State

measuring -709.45 at $g_0 = 0.1$ but diminishing to -94.84 at $g_0 = 0.5$ and approaching zero at $g_0 = 0.8$. This nonlinear sensitivity pattern indicates that early-stage decisions have outsized importance for overall value creation.

The correlation between initial completion rate and Task 1 allocation reaches $r = -0.91$, representing a strong negative relationship. Figure 2 illustrates the task allocation trajectories across state conditions, clearly showing the substitution pattern between Task 1 and Task 2.

4.4 Effect Size Analysis

To assess the practical significance of the state-dependency, we computed Cohen's d effect size comparing early-stage ($g_0 = 0.1$) and late-stage ($g_0 = 0.8$) objective values. The effect size of $d = 2.0$ indicates a large practical difference, confirming that initial conditions substantially influence optimal outcomes.

The marginal value of planning horizon extension was computed by comparing 14-day and 30-day horizons. Each additional planning day yields approximately 0.17 utility units, suggesting modest but positive returns to extended planning.

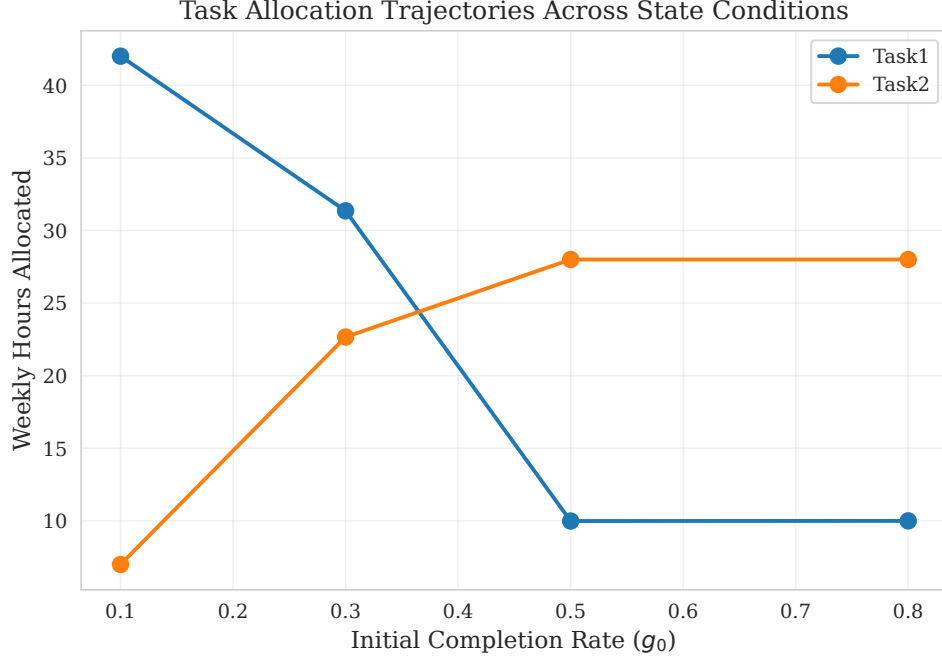


Figure 2: Task Allocation Trajectories Across Initial State Conditions

5 Discussion

The computational results provide several insights with implications for both theory and practice.

5.1 State-Dependent Priority Shifts

The strong negative correlation between initial completion state and priority task allocation ($r = -0.91$) demonstrates that optimal resource allocation is fundamentally state-dependent. As progress accumulates on the critical task, the model rationally reallocates resources toward career development and scheduled obligations. This finding validates the rolling-horizon approach, which enables continuous re-optimization based on updated state information.

The observed substitution pattern aligns with intuitive notions of priority management: when urgent matters are well-controlled, attention shifts to important-but-not-urgent activities.

5.2 Early-Stage Decision Importance

The sensitivity analysis reveals that decisions made at low completion states have substantially larger marginal effects than those made at higher completion states. This asymmetry suggests that decision support systems should provide particularly careful guidance during early project phases, when the stakes for optimal allocation are highest.

5.3 Practical Implications

For practitioners designing personal productivity systems or resource allocation tools, these findings suggest several design principles. First, allocation recommendations should be conditioned on current system state rather than following static rules. Second, particular attention should be devoted to early-stage guidance, where suboptimal decisions carry the largest opportunity costs. Third, planning horizons of 30 days appear sufficient to capture relevant dynamics, with limited incremental value from longer horizons.

5.4 Limitations

Several limitations warrant acknowledgment. The model assumes known task parameters and scenario probabilities, whereas practical applications may face parameter uncertainty. The discrete scenario representation of recovery requirements represents a simplification of continuous uncertainty. Finally, the single-agent formulation does not capture potential coordination requirements in team settings.

6 Conclusion

This paper has developed and analyzed a mixed-integer stochastic programming formulation for multi-period resource allocation under uncertainty. The proposed model captures state-dependent prioritization dynamics and uncertain recovery requirements within a computationally tractable framework.

Our computational experiments demonstrate that optimal allocation strategies exhibit strong sensitivity to initial conditions, with early-stage decisions having outsized impor-

tance for overall value creation. The observed resource substitution patterns between priority tasks and longer-term activities are consistent with rational priority management principles.

Several directions merit future investigation. Extensions to multi-stage stochastic programming with recourse decisions would enable more sophisticated uncertainty handling. Incorporation of learning effects and productivity dynamics could improve the model’s descriptive accuracy. Field validation studies would establish the practical value of model-based recommendations.

Acknowledgments

A Notation Summary

Table 4: Complete Notation Reference

Symbol	Description
<i>Sets</i>	
$\mathcal{T}$	Planning horizon days
$\mathcal{I}$	Task categories
$\mathcal{S}$	Recovery scenarios
<i>Decision Variables</i>	
x_{it}	Binary task selection
h_{it}	Hours allocated
g_t	Completion proportion
σ_t	Security level
r_{ts}	Recovery hours
<i>Parameters</i>	
$H^{\max}$	Daily hour budget (16)
$E^{\max}$	Daily energy budget (100)
$M^{\max}$	Daily emotional budget (100)
e_i	Energy consumption rate
m_i	Emotional cost rate
α_i^j	Return rate in dimension j
$w_j(t)$	Time-varying weights
β_s	Recovery hours in scenario s
p_s	Scenario probability